Table 10.—Source Code File Format

Field name	Type	Blank	Description
Src_Cd	A 2*	N	2-digit code.
SrcCd_Desc	A 60	N	Description of source code that identifies the type of nutrient data.

* Primary key for the Source Code file.

Data Derivation Code Description File (file name = DERIV_CD). This file (Table 11) provides information on how the nutrient values were determined. The file contains the derivation codes and their descriptions.

- Links to the Nutrient Data file by Deriv_Cd

Table 11.—Data Derivation Code File Format

Field name	Type	Blank	Description
Deriv_Cd	A 4*	N	Derivation Code.
Deriv_Desc	A 120	N	Description of derivation code giving specific information on how the value was determined.

* Primary key for the Data Derivation Code file.

For example, the data derivation code that indicates how α-tocopherol (Nutrient No. 323) in Emu, fan fillet, raw (NDB. No. 05623) was calculated is BFSN. The breakdown of the code is as follows:

B = based on another form of the food or a similar food;
F = concentration adjustment used;
S = solids, the specific concentration adjustment used; and
N = retention factors not used.

The Ref_NDB_No is 05621 Emu, ground, raw. This means that the analytical α-tocopherol value in the total solids of emu, ground, raw is used to calculate the α-tocopherol in the total solids of emu, fan fillet, raw.

$$N_t = (N_s * S_s)/S_t$$

where:
N_t = the nutrient content of the target item,
N_s = the nutrient content of the source item,
For NDB No. 05621, α-tocopherol = 0.24 mg/100 g
S_s = the total solids content of the source item, and
For NDB No. 05621, solids = 27.13 g/100 g

S_t = the total solids content of the target item.
For NDB No. 05623, solids = 2538 g/100 g

So, using this formula for the above example:

N_t = (0.24 × 25.38)/27.13 = 0.22 mg/100 g α-tocopherol in Emu, fan fillet, raw

Only items that were imputed starting with SR14 (2001) will have both derivation codes and reference NDB numbers. Other items that have been imputed outside the NDBS will have data derivation codes, but the Ref_NDB_No field will be blank.

Weight File (file name = WEIGHT). This file (Table 12) contains the weight in grams of a number of common measures for each food item.

- Links to Food Description file by NDB_No
- Links to Nutrient Data file by NDB_No

Table 12.— Weight File Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	5-digit Nutrient Databank number.
Seq	A 2*	N	Sequence number.
Amount	N 5.3	N	Unit modifier (for example, 1 in “1 cup”).
Msre_Desc	A 84	N	Description (for example, cup, diced, and 1-inch pieces).
Gm_Wgt	N 7.1	N	Gram weight.
Num_Data_Pts	N 3	Y	Number of data points.
Std_Dev	N 7.3	Y	Standard deviation.

* Primary key for the Weight file.

Footnote File (file name = FOOTNOTE)**.** This file (Table 13) contains additional information about the food item, household weight, and nutrient value.

- Links to the Food Description file by NDB_No
- Links to the Nutrient Data file by NDB_No and when applicable, Nutr_No
- Links to the Nutrient Definition file by Nutr_No, when applicable

Table 13.—Footnote File Format

Field name	Type	Blank	Description
NDB_No	A 5	N	5-digit Nutrient Databank number.
Footnt_No	A 4	N	Sequence number. If a given footnote applies to more than one nutrient number, the same footnote number is used. As a result, this file cannot be indexed.
Footnt_Typ	A 1	N	Type of footnote: D = footnote adding information to the food description; M = footnote adding information to measure description; N = footnote providing additional information on a nutrient value. If the Footnt_typ = N, the Nutr_No will also be filled in.
Nutr_No	A 3	Y	Unique 3-digit identifier code for a nutrient to which footnote applies.
Footnt_Txt	A 200	N	Footnote text.

Sources of Data Link File (file name = DATSRCLN). This file (Table 14) is used to link the Nutrient Data file with the Sources of Data table. It is needed to resolve the many-to-many relationship between the two tables.

- Links to the Nutrient Data file by NDB No. and Nutr_No
- Links to the Nutrient Definition file by Nutr_No
- Links to the Sources of Data file by DataSrc_ID

Table 14.—Sources of Data Link File Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	5-digit Nutrient Databank number.
Nutr_No	A 3*	N	Unique 3-digit identifier code for a nutrient.
DataSrc_ID	A 6*	N	Unique ID identifying the reference/source.

* Primary key for the Sources of Data Link file.

Sources of Data File (file name = DATA_SRC). This file (Table 15) provides a citation to the DataSrc_ID in the Sources of Data Link file.

- Links to Nutrient Data file by NDB No. through the Sources of Data Link file

Table 15.—Sources of Data File Format

Field name	Type	Blank	Description
DataSrc_ID	A 6*	N	Unique number identifying the reference/source.
Authors	A 255	Y	List of authors for a journal article or name of sponsoring organization for other documents.
Title	A 255	N	Title of article or name of document, such as a report from a company or trade association.
Year	A 4	Y	Year article or document was published.
Journal	A 135	Y	Name of the journal in which the article was published.
Vol_City	A 16	Y	Volume number for journal articles, books, or reports; city where sponsoring organization is located.
Issue_State	A 5	Y	Issue number for journal article; State where the sponsoring organization is located.
Start_Page	A 5	Y	Starting page number of article/document.
End_Page	A 5	Y	Ending page number of article/document.

* Primary key for the Sources of Data file.

Abbreviated File

The Abbreviated file **(**file name = ABBREV) is available in ASCII format and as a Microsoft Excel spreadsheet. It contains all the food items found in the relational database, but with fewer nutrients and other related information. The abbreviated file does not include values for starch, fluoride, betaine, vitamin D_2 and D_3, added vitamin E, added vitamin B_{12}, alcohol, caffeine, theobromine, phytosterols, individual amino acids, individual fatty acids, or sugars. Table 16 lists all the nutrients included in the abbreviated file. Starting with SR22 (2009), Vitamin D in µg and IU was added to the Abbreviated file. The ASCII file (Table 16) is in delimited format. Fields are separated by a caret (^) and text fields are surrounded by tildes (~). Data refer to 100 g of the edible portion of the food item. Decimal points are included in the fields. Missing values are denoted by the null value of two consecutive carets (^^) or two carets and two tildes (~~). The file is sorted in ascending order by the NDB number. Two common measures are provided, which are the first two common measures in the Weight file for each NDB number. To obtain values per one of the common measures, multiply the value in the desired nutrient field by the value in the desired common measure field and divide by 100. For example, to calculate the amount of fat in 1 tablespoon of butter (NDB No. 01001):

V_H=(N*CM)/100

where:
V_H = the nutrient content per the desired common measure,
N = the nutrient content per 100 g,
For NDB No. 01001, fat = 81.11 g/100 g
CM = grams of the common measure.
For NDB No. 01001, 1 tablespoon = 14.2 g

So using this formula for the above example:

V_H = (81.11*14.2)/100 = 11.52 g fat in 1 tablespoon of butter

This file is a flat file and is provided for those users who do not need a relational database. It contains the information in one record per food item and is suitable for importing into a spreadsheet. The data file has been imported into a Microsoft Excel 2007 spreadsheet for users of that application. Users of other software applications can import either the Microsoft Excel 2007 spreadsheet or the ASCII files. If additional information is needed, this file can be linked to the other SR files by the NDB number.

Table 16.—Abbreviated File Format

Field name	Type	Description
NDB_No.	A 5*	5-digit Nutrient Databank number.
Shrt_Desc	A 60	60-character abbreviated description of food item.†
Water	N 10.2	Water (g/100 g)
Energ_Kcal	N 10	Food energy (kcal/100 g)
Protein	N 10.2	Protein (g/100 g)
Lipid_Tot	N 10.2	Total lipid (fat)(g/100 g)
Ash	N 10.2	Ash (g/100 g)
Carbohydrt	N 10.2	Carbohydrate, by difference (g/100 g)
Fiber_TD	N 10.1	Total dietary fiber (g/100 g)
Sugar_Tot	N 10.2	Total sugars (g/100 g)
Calcium	N 10	Calcium (mg/100 g)
Iron	N 10.2	Iron (mg/100 g)
Magnesium	N 10	Magnesium (mg/100 g)
Phosphorus	N 10	Phosphorus (mg/100 g)
Potassium	N 10	Potassium (mg/100 g)
Sodium	N 10	Sodium (mg/100 g)

Field name	Type	Description
Zinc	N 10.2	Zinc (mg/100 g)
Copper	N 10.3	Copper (mg/100 g)
Manganese	N 10.3	Manganese (mg/100 g)
Selenium	N 10.1	Selenium (µg/100 g)
Vit_C	N 10.1	Vitamin C (mg/100 g)
Thiamin	N 10.3	Thiamin (mg/100 g)
Riboflavin	N 10.3	Riboflavin (mg/100 g)
Niacin	N 10.3	Niacin (mg/100 g)
Panto_acid	N 10.3	Pantothenic acid (mg/100 g)
Vit_B6	N 10.3	Vitamin B_6 (mg/100 g)
Folate_Tot	N 10	Folate, total (µg/100 g)
Folic_acid	N 10	Folic acid (µg/100 g)
Food_Folate	N 10	Food folate (µg/100 g)
Folate_DFE	N 10	Folate (µg dietary folate equivalents/100 g)
Choline_Tot	N 10	Choline, total (mg/100 g)
Vit_B12	N 10.2	Vitamin B_{12} (µg/100 g)
Vit_A_IU	N 10	Vitamin A (IU/100 g)
Vit_A_RAE	N 10	Vitamin A (µg retinol activity equivalents/100g)
Retinol	N 10	Retinol (µg/100 g)
Alpha_Carot	N 10	Alpha-carotene (µg/100 g)
Beta_Carot	N 10	Beta-carotene (µg/100 g)
Beta_Crypt	N 10	Beta-cryptoxanthin (µg/100 g)
Lycopene	N 10	Lycopene (µg/100 g)
Lut+Zea	N 10	Lutein+zeazanthin (µg/100 g)
Vit_E	N 10.2	Vitamin E (alpha-tocopherol) (mg/100 g)
Vit_D_mcg	N 10.1	Vitamin D (µg/100 g)
Vit_D_IU	N 10	Vitamin D (IU/100 g)
Vit_K	N 10.1	Vitamin K (phylloquinone) (µg/100 g)
FA_Sat	N 10.3	Saturated fatty acid (g/100 g)
FA_Mono	N 10.3	Monounsaturated fatty acids (g/100 g)
FA_Poly	N 10.3	Polyunsaturated fatty acids (g/100 g)

Field name	Type	Description
Cholestrl	N 10.3	Cholesterol (mg/100 g)
GmWt_1	N 9.2	First household weight for this item from the Weight file.‡
GmWt_Desc1	A 120	Description of household weight number 1.
GmWt_2	N 9.2	Second household weight for this item from the Weight file.‡
GmWt_Desc2	A 120	Description of household weight number 2.
Refuse_Pct	N 2	Percent refuse.§

* Primary key for the Abbreviated file.
† For a 200-character description and other descriptive information, link to the Food Description file.
‡ For the complete list and description of the measure, link to the Weight file.
§ For a description of refuse, link to the Food Description file.

Update Files

The update files contain changes made between the last release, SR26 (2013), and the current release, SR27 (2014). Update files in ASCII are provided for those users who reformatted previous releases for their systems and wish to do their own updates. If a release earlier than SR26 is used, it is necessary to first obtain the update files for that release through SR26, update the database to SR26, and then use the update files provided with SR27. Update files from previous releases are available on NDL's web site: http://www.ars.usda.gov/nutrientdata.

New data added to SR27 are given in the following files:
- ADD_FOOD for descriptions of the new items (307 records);
- ADD_NUTR for added nutrient data (30,317 records);
- ADD_WGT for added weight and measure data (398 records); and
- ADD_FTNT for added footnotes (14 records).

These files are in the same formats as the Food Description file, the Nutrient Data file, the Weight file, and the Footnote file.

Five files contain changes made since SR26 (2013):
- CHG_FOOD contains records with changes in the descriptive information for a food item (310 records);
- CHG_NUTR contains changes to the following fields: nutrient values, standard errors, number of data points, source code, and data derivation code (20,278 records);

- CHG_WGT contains records with changes to the gram weights or measure information (300 records); and
- CHG_FTNT contains records with changes to footnotes (10 records).
- CHG_NDEF contains records with changes to the nutrient definitions (1 record)

If the values in any fields have changed, the entire record is included for that file. These files are in the same format as the Food Description, Nutrient Data, Weight, Footnote, and Nutrient Definition files.

Four files contain records that were deleted since SR26 (2013):
- DEL_FOOD file (Table 17) lists those food items that were deleted from the database (152 records);
- DEL_NUTR file (Table 18) lists those nutrient values that were removed from the database (8,639 records);
- DEL_WGT contains any gram weights that were removed (307 records). These records are in the same format as the Weight file (Table 12); and
- DEL_FTNT contains any footnotes that were removed from the database (Table 19). Starting with SR19, if a given footnote applied to more than one nutrient number, the same footnote number can be used. When these footnote numbers are updated, the extra footnotes are deleted (27 records).

Table 17.—Foods Deleted Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	Unique 5-digit number identifying deleted item.
Shrt_Desc	A 60	N	60-character abbreviated description of food item.

* Primary key for Foods Deleted file.

Table 18. —Nutrients Deleted Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	Unique 5-digit number identifying the item that contains the deleted nutrient record.
Nutr_No	A 3	N	Nutrient number of deleted record.

* Primary key for Nutrients Deleted file.

Table 19.—Footnotes Deleted Format

Field name	Type	Blank	Description
NDB_No	A 5*	N	Unique 5-digit number identifying the item that contains the deleted nutrient record.
Footnt_No	A 4	N	Sequence number.
Footnt_Typ	A 1	N	Type of footnote of deleted record.

* Primary key for Footnotes Deleted file.

Update files in ASCII are also provided for the Abbreviated file:

- CHG_ABBR file contains records for food items where a food description, household weight, refuse value, or nutrient value have been added, changed, or deleted since SR26. This file is in the same format as the Abbreviated file (Table 16);
- DEL_ABBR contains food items that have been removed from the database; it is in the same format as DEL_FOOD; and
- ADD_ABBR contains food items added since SR26; it is also in the same format as the Abbreviated file.

Summary

A number of food items have been added to the database using new data from NFNAP, the food industry, and other sources. Other foods have had nutrient values updates. In particular, the sodium content of those foods which are major contributors of sodium to the diet—primarily commercially processed and restaurant foods— has been targeted for nutrient analysis. A number of food items, no longer on the market, such as certain processed foods, have been removed. These are described in “Specific Changes for SR27” (p. 2). The next release, SR28, available in 2015, will contain additional items and updates.

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